

# Matthew J. Nitzken

Principal Machine Learning Engineer

Remote - Louisville, KY

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## About

I'm a machine learning engineer and data scientist with a background in electrical, computer, and bioengineering, and more than 15 years working in AI. I'm currently a Principal Machine Learning Engineer at DataRobot, where I lead the engineering behind our LLM agents and agentic workflows and build generative AI products. It's been the near-total focus of my recent work, and the direction I want to keep building in. The rest of my career has been full-stack machine learning and data science in large-scale data environments, not just building models but turning new ideas into real, shipped products, and I gravitate toward hard problems that haven't been solved yet and getting them into production. Along the way I've worked across generative AI, LLMs, and agents; time series and deep learning; healthcare; data engineering; medical diagnostics and research; image and signal processing; technical publishing; software development; and controls automation. I enjoy collaborating with, learning from, and mentoring the people around me.

## Skills

|                          |                               |                                |                                    |
|--------------------------|-------------------------------|--------------------------------|------------------------------------|
| <b>Python</b>            | <b>Machine Learning</b>       | <b>Deep Learning</b>           | <b>Time Series</b>                 |
| <b>LLMs &amp; Agents</b> | <b>Generative AI</b>          | <b>Model Explainability</b>    | <b>Big Data</b>                    |
| <b>Data Engineering</b>  | <b>Cloud &amp; Kubernetes</b> | <b>Test-Driven Development</b> | <b>Mentorship &amp; Leadership</b> |

## Education

### University of Louisville

Doctor of Philosophy (Ph.D.), in Electrical and Computer Engineering

[Louisville, Kentucky](#)

2010 – 2015

- Dissertation: Shape Analysis of the Human Brain

### University of Louisville

Masters of Engineering (M.Eng.), in Bioengineering

[Louisville, Kentucky](#)

2009 – 2010

### University of Louisville

Bachelor of Science (B.S.), in Bioengineering

[Louisville, Kentucky](#)

2005 – 2009

## Experience

### DataRobot

[Louisville, KY — Remote](#)

Data Scientist → Senior → Staff → Principal Machine Learning Engineer

Jul 2019 – Present

Engineering lead for DataRobot's LLM agent and agentic workflow products, after growing from data scientist across time series, model explainability, platform architecture, and generative AI.

- Lead the engineering for DataRobot's LLM agents and agentic workflows, and build the generative AI and multi-agent products — including development templates that let teams run frameworks like LangGraph, CrewAI, and LlamaIndex inside DataRobot.
- Co-architected af-components, a framework for building agentic generative AI solutions: drop-in templates carry their own Terraform/Pulumi infrastructure so a full solution deploys in one command, and central fixes for bugs, CVEs, and features push out to internal and external users automatically.
- Joined the in-flight Universal SHAP effort and carried it to delivery: because SHAP doesn't natively work on time series data, developed and validated a new method to make it work, then led the engineering to ship it — putting SHAP explanations on 200+ model blueprints that had none, across DataRobot SaaS (single- and multi-tenant) and on-prem.
- Reworked a stalled plan to break up a core platform monolith into one that could actually be implemented — restructuring test infrastructure, splitting code into new repositories with their own CI/CD, and reworking core code; the resulting approach was picked up and reused across other teams.

- Created and led several core time series products — incremental (online) learning that removed training-size limits, Segmented Modeling for large intermixed datasets, and an unsupervised clustering framework spanning feature processing, modeling, insights, and predictions.
- Built DataRobot's early time series trust and explainability features — feature importance, feature impact, and drift — adapting AutoML methods into then-unexplored territory, on top of large-scale data-warehousing and feature-engineering infrastructure I designed.
- *Experience focused on Python, LLMs, LangGraph, TensorFlow, Keras, Docker, Kubernetes, Helm, AWS, GCP, Snowflake, Postgres, MongoDB, and TDD*

## Humana

*Louisville, KY / Boston, MA*

**Analytics Consultant → Lead Data Scientist**

*Oct 2016 – Jul 2019*

Built and led machine learning across Humana's clinical and Medicare STARS analytics, moving from analytics consultant to lead data scientist across three teams.

- Replaced Humana's Medicare STARS rating forecasts — previously unreliable ARIMA projections — with Monte Carlo simulations (one per plan and measure, plus a global model) that produced a probability curve across the 1–5 STARS scale, showing leadership where each plan was likely to land, its odds of slipping, and where to focus money and resources.
- Built a large set of models predicting member health trajectories across conditions common to Medicare members (diabetes, COPD, flu vaccination), ranging from straightforward ML to SAS/Spark-fronted pipelines to TensorFlow deep learning.
- Later joined a new Boston team building real-time deep learning for Medicare members — migrating data off on-prem to GCP and building the data pipelines and front-end-backed applications around the models.
- *Experience using Python, TensorFlow, Keras, SkLearn, SAS, GCP, AWS, Hadoop, SQL, Airflow, and TDD*

## Alliant Technologies

*Louisville, KY*

**Software Engineer → Lead Software and Controls Engineer**

*May 2015 – Oct 2016*

Software and controls developer for automated high-speed sorter systems, promoted to lead the software development and commissioning teams.

- Led software development and controls commissioning through build-out and QA on large-scale material-handling projects.
- Built an Ethernet/IP communications stack for high-speed sortation, and a Python server that simulated the control server so new systems could be tested before deployment.

## NeuroAtlas

*Louisville, KY*

**Founder and Machine Learning Engineer**

*Mar 2013 – May 2015*

Co-founder and lead machine learning engineer at NeuroAtlas, building neurological diagnostic software for autism spectrum disorders.

- Went through two healthcare accelerators (XLerateHealth, LaunchIT) and ran several grant-funded initiatives; co-led direction, design, and fundraising.

## University of Louisville

*Louisville, KY*

**Graduate Researcher — Bioluminescence Laboratory**

*Jul 2010 – May 2015*

Ph.D. research developing machine learning for digital diagnostics.

- Developed and published 2D/3D shape, image-, and signal-processing algorithms and computer-aided diagnostic ML, across autism, dyslexia, cancer, Alzheimer's, and cardiac/EMG/ECG signals.
- Presented research at national and international conferences, through talks and posters.

## Publications and Patents

Please see [publications.mattnitzken.com](http://publications.mattnitzken.com) for an up-to-date list of my publications and patents.